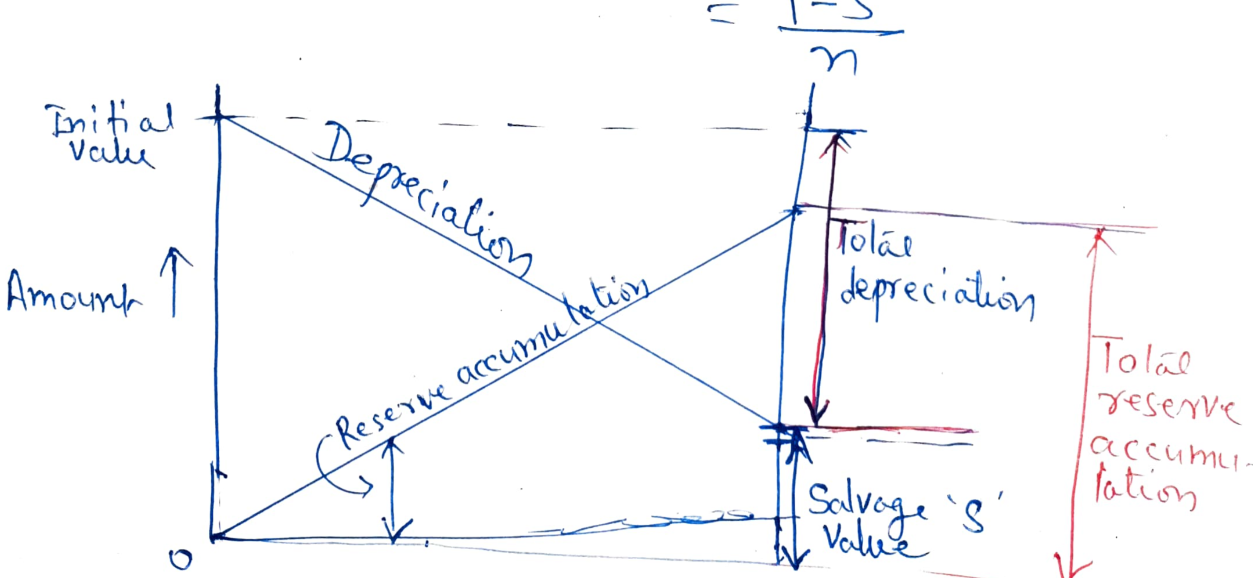
Depreciation :-

(i) Straight line Method :- Certain depreciation occurs according to straight line law.

A constant depreciation charge is made every year on the basis of total depreciation (initial cost - Salvage Value). Salvage Value (scrap Value) is priori known as some percentage of initial value ~~over~~ after the period of useful life.

$$\text{Annual depreciation charge} = \frac{(\text{Initial cost} - \text{Salvage Value})}{\text{No. of years of useful life}}$$

$$= \frac{P-S}{n}$$



This method neglect intrest that depreciation reserve would earn.

(ii) SINKING FUND METHOD:- In this method depreciation changes set aside each year and invested at intrest rate compounded annually. This amount will be equal to replacement cost of the installation at the end of its useful life.

As compare to straight line method, it requires smaller annual amounts.

let $P \rightarrow$ Initial Value
$n \rightarrow$ useful life of the equipment
$S \Rightarrow$ Scrap or salvage Value
$r \rightarrow$ rate of intrest /annum.

If annual deposit be 'q', it will earn intrest 'rq' in one year.

At the end of one year it will worth $q + qr = q(1+r)$
" " " n^{th} " " " $= q(1+r)^n$

Thus an amount of 'q' deposited at the end of first year, in $(n-1)$ years will be $= q(1+r)^{n-1}$

An amount of q deposited at the end of second year in $(n-2)$ years will be $= q(1+r)^{n-2}$

Similarly an amount 'q' deposited at the end of 3^{rd} year in $(n-3)$ years will be $= q(1+r)^{n-3}$

" — " " "

$(n-1)^{th}$ year in in one year will be $= q(1+r)$

Thus after n years total sinking fund will be

$$= q\left[(1+r)^{n} + (1+r)^{n-1} + (1+r)^{n-2} + (1+r)^{n-3} + \dots + (1+r)\right]$$

The above terms are in <u>geometrical progression</u>

$$\text{Total sum} = \frac{\left(\frac{1}{1+r}\right) q - (1+r)^{n-1}}{\frac{1}{1+r} - 1} = \frac{1-(1+r)(1+r)^{n-1}}{1-(1+r)}$$

$$= q\,\frac{(1+r)^{n}-1}{r}$$

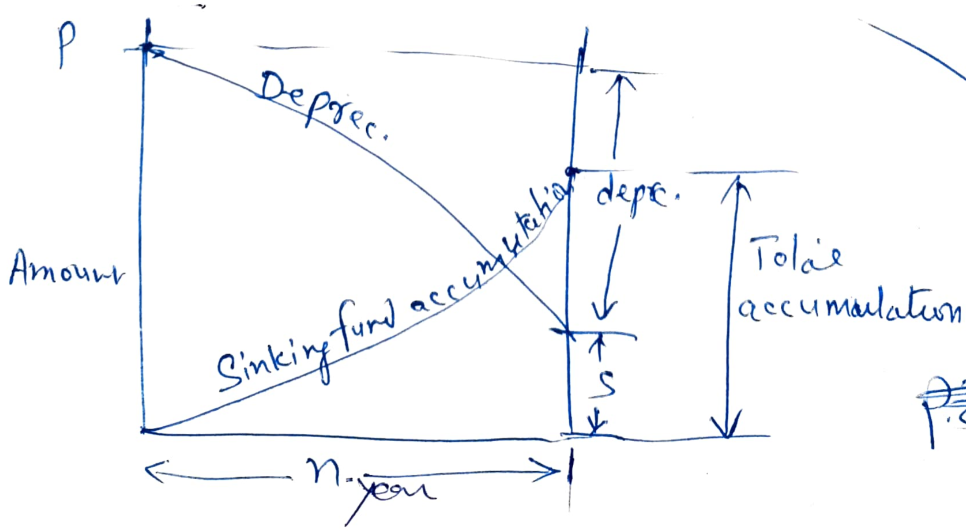
This sum must be equal to the cost of replacement Q which equal to $(P-S)$

$$Q = (P-S) = \text{Reserve accumulation}$$

$$Q = q\,\frac{(1+r)^{n}-1}{r}$$

or annual deposit $q = \dfrac{Qr}{(1+r)^{n}-1}$

Annual sinking fund depreciation reserve. $\boxed{q = (P-S)\left[\frac{r}{(1+r)^{n}-1}\right]}$



P.T.O.

(iii) Fixed Percentage Method:
OR (Declining balance Method)

In this method the depreciation ~~reserve~~ during any year is a fixed percentage 'x' of the remaining balance of initial investment minus total accumulated depreciation at the begining of the year.

Let C → initial investment

n → useful life in years

Annual depreciation during first year $= \frac{Cx}{100}$

Remaining balance at the begining of second year $= C - \frac{Cx}{100}$

$$= C\left(1 - \frac{x}{100}\right)$$

Annual depreciation during second yr $= C\left(1-\frac{x}{100}\right)\frac{x}{100}$

Remaining balance at begining of third yr. $= C\left(1-\frac{x}{100}\right) - C\left(1-\frac{x}{100}\right)\frac{x}{100}$

$$= C\left(1-\frac{x}{100}\right)^2$$

Annual depreciation during 3rd year $= C\left(1-\frac{x}{100}\right)^2\frac{x}{100}$

Remaining balance at beg. 4th year $= C\left(1-\frac{x}{100}\right)^2 - C\left(1-\frac{x}{100}\right)^2\frac{x}{100}$

$$= C\left(1-\frac{x}{100}\right)^3$$

Remaining balance at the end of n yrs (Life of equipment)

$$= c\left(1-\frac{x}{100}\right)^n$$

The above balance at the end of n year should be equal to Salvage Value 'S'

Thus $\boxed{c\left(1-\frac{x}{100}\right)^n = S}$

c, S, n are known

x can be found.